

The Command Line

Introduction

3 pieces to a terminal

- terminal emulator
 - only handles text, input, colors, and scrolling
 - does not know what the commands mean
- Shell
 - reads and interprets the command given
- Operating System (Kernal)
 - The shell asks the OS to:
 - create processes, allocate memory, access files, and communicate with hardware

Introduction (cont.)

Main operation loop: Input -> Interpret -> Execute -> Output

The terminal is very powerful because:

- All the commands give you precise control over the computer
- You can create scripts to automate processes
- Having no UI increases the speed
- Have access to SSH

This is where the terminal excels over gui's

How Shells Parse Commands

The shell does not just split the command by spaces, it goes through multiple more phases:

1. Lexing or Tokenization

- i. tokens are separated by spaces, recognizes special characters, recognizes quotes

How Shells Parse Commands (cont.)

2. Parsing

- i. In this step a syntax tree is made
- ii. The grammar is seen as:
 - a. command, pipeline, redirect, conditional execution, subshells
 - b. ex:

```
redirect(  pipeline(          command(echo, $HOME),  
command(grep, user)    ),    out.txt  )
```
 - c. This Structure is referred to an Abstract Syntax Tree

How Shells Parse Commands (cont.)

3. Expansions

i. The shell performs expansions before it executes in this order:

a. brace expansion: `echo file{1..3}` -> echos files 1 - 3

b. Tilde expansion: `echo ~` -> `echo /home/you`

c. parameter expansion: `echo $HOME` -> echos the HOME variable

d. command substitution: `echo $(date)` -> echos the result of the command date

e. arithmetic expansion: `echo $((2 + 3))` -> echos 5

f. Word Splitting: Splits words by spaces, unless quoted

g. globbing, or filename expansion: `echo *.txt` echos all files that end in .txt

How Shells Parse Commands (cont.)

4. Redirections

- i. Before execution, the shell sets up file descriptors

5. Execution

- i. decides whether the command is a function, built-in, or external
- ii. runs

